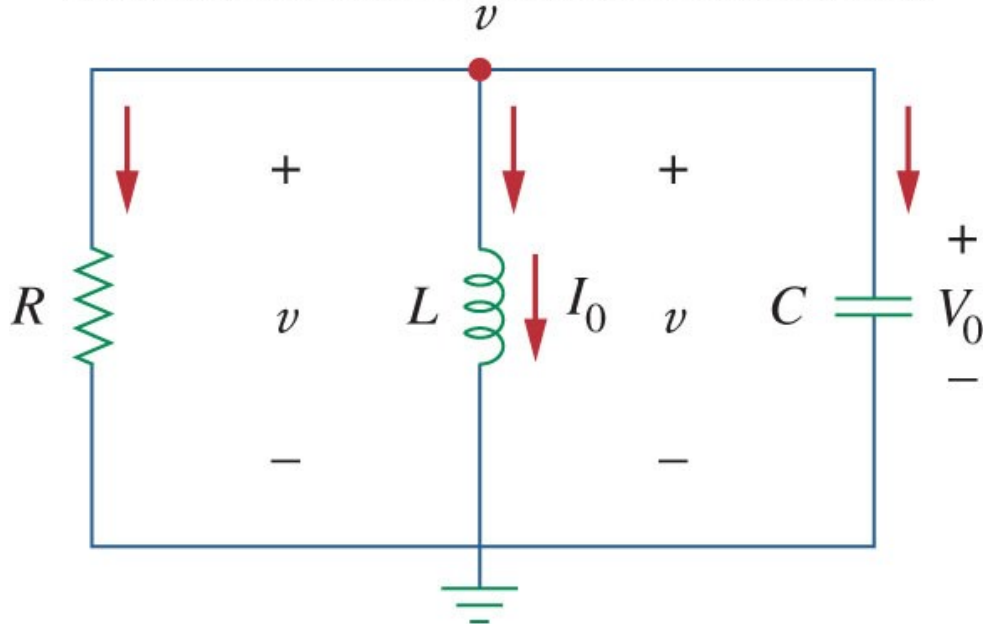




Source-Free Parallel RLC Network

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Source-Free Parallel RLC Network - $v(t)$

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

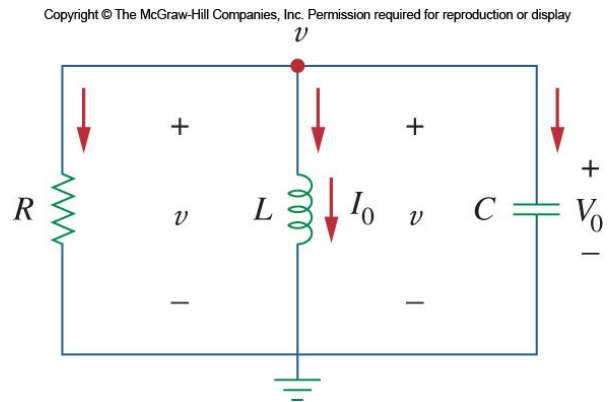
- The characteristic equation is:

$$s^2 + \frac{1}{RC} s + \frac{1}{LC} = 0$$

From this, we can find the roots of the characteristic equation to be:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$
$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Again, there are three scenarios to consider.





Three Damping Cases - $v(t)$

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$
$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- For the overdamped case, the roots are real and negative,

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- For critically damped, the roots are real and equal

$$v(t) = (A_1 t + A_2) e^{-\alpha t}$$

- For the underdamped case, the roots are complex

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$



Three Damping Cases - $i(t)$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$
$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- For the overdamped case, the roots are real and negative,

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- For critically damped, the roots are real and equal

$$i(t) = (A_1 t + A_2) e^{-\alpha t}$$

- In the underdamped case, the roots are complex

$$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

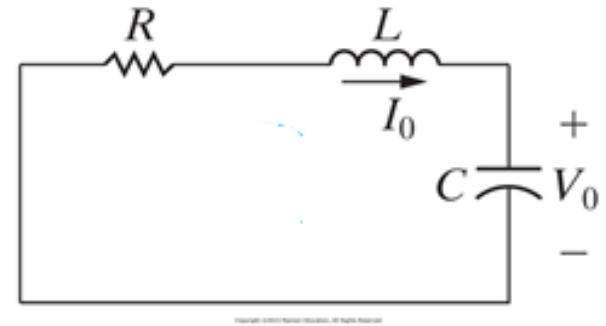
Series vs. Parallel (Source-Free RLC Circuit)

• Series $\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(t) = (A_1 t + A_2) e^{-\alpha t}$$

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

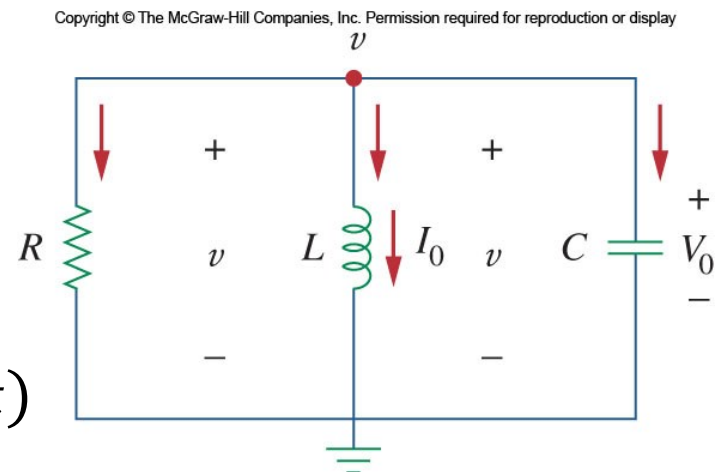


• Parallel $\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(t) = (A_1 t + A_2) e^{-\alpha t}$$

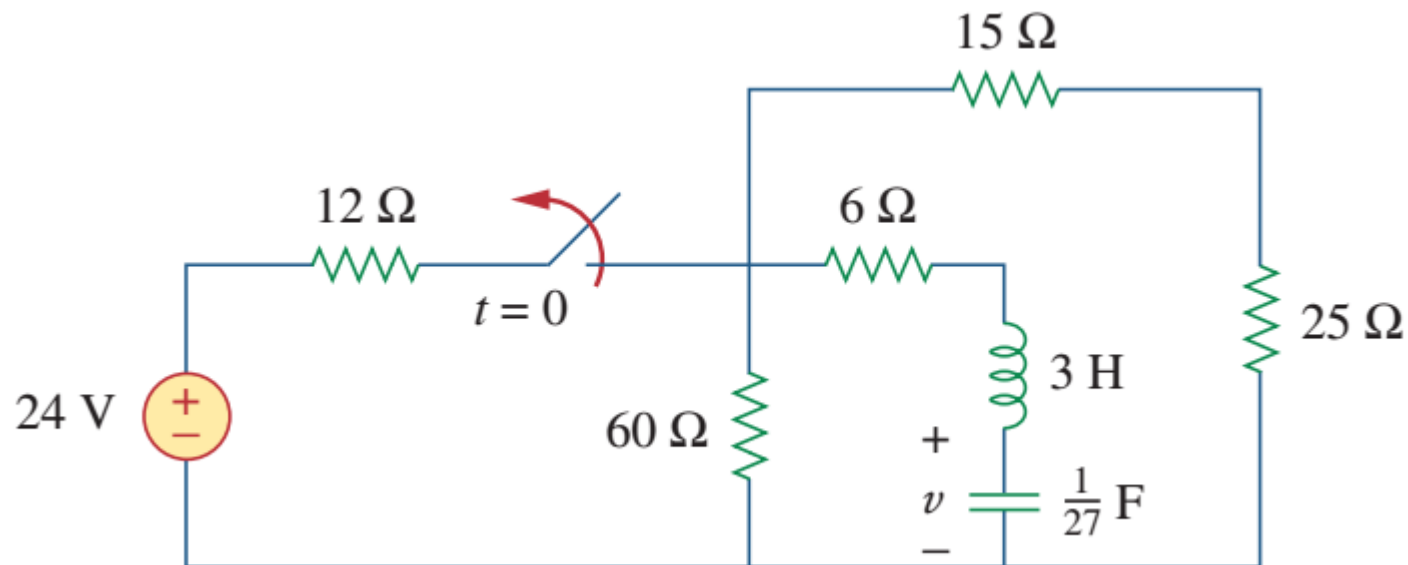
$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$





Practice

- Find $v(t)$ for $t > 0$



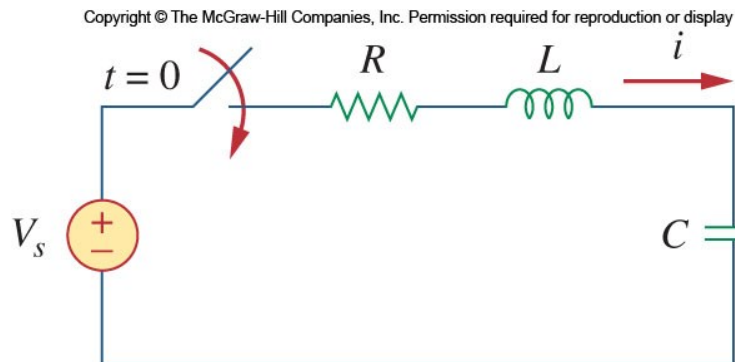


Outline

- Natural Response Series/Parallel RLC circuit
Source-free
- Step Response of a Series/Parallel RLC Circuit
With Independent Source
- General 2nd-order circuits



Step Response of a Series RLC Circuit



$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

- The solution

$$v(t) = v_t(t) + v_{ss}(t)$$

- The complete solutions for the three conditions of damping are:

$$v(t) = V_s + (A_1 e^{s_1 t} + A_2 e^{s_2 t}) \quad (\text{Overdamped})$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically Damped})$$

$$v(t) = V_s + (B_1 \cos \omega_d t + B_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$





Example

- Find $v(t)$ and $i(t)$ for $t > 0$.

Consider three cases:

- $R = 5\Omega$
- $R = 4\Omega$
- $R = 1\Omega$

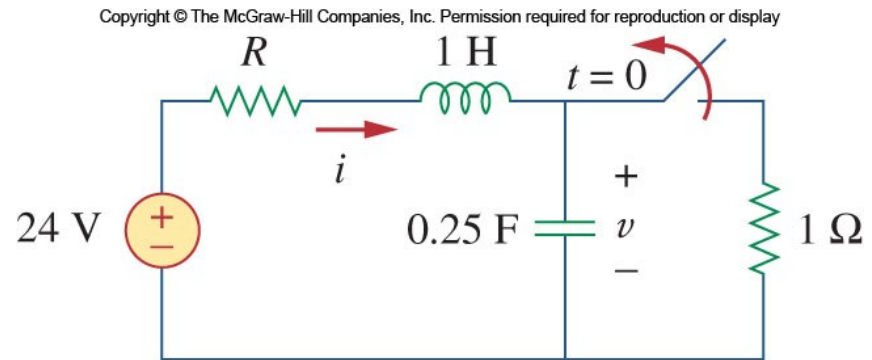
When $R = 5\Omega$,

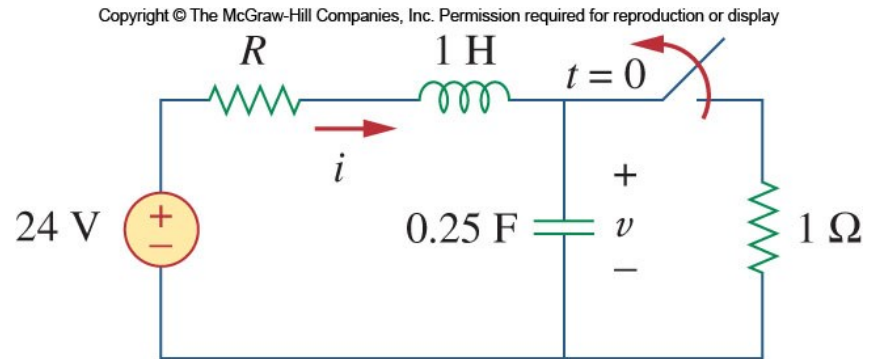
- For $t < 0$, switch closed, capacitor open, inductor shorted.
- For $t > 0$, switch open, a series RLC network

- $v(0) = 4V \quad i(0) = 4A = C \frac{dv(0)}{dt}, \quad \frac{dv(0)}{dt} = 16$

$$\alpha = \frac{R}{2L} = 2.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 2, \quad s_{1,2} = -1, -4 \quad \text{Overdamped.}$$

$$v(t) = v_s + (A_1 e^{-t} + A_2 e^{-4t})$$



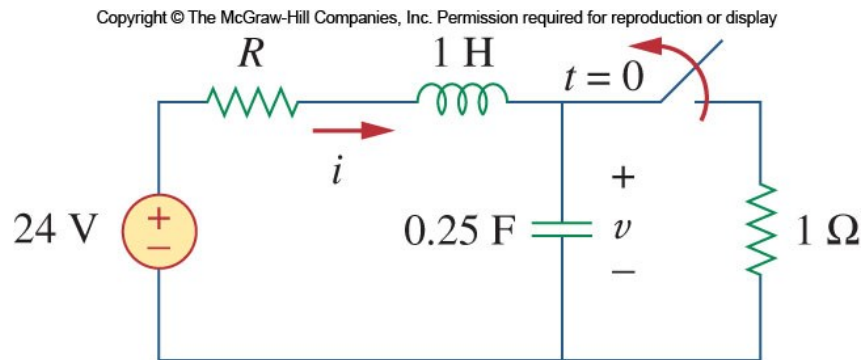


When $R = 4\Omega$,

- For $t < 0$, switch closed, capacitor open, inductor shorted.
- For $t > 0$, switch open, a series RLC network
- $v(0) = 4.8V, i(0) = 4.8A = C \frac{dv(0)}{dt}, \frac{dv(0)}{dt} = 19.2$

$$\alpha = \frac{R}{2L} = 2, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 2, \quad s_{1,2} = -2 \quad \text{Critically damped}$$

$$v(t) = v_s + (A_1 + A_2 t)e^{-2t}$$



When $R = 1\Omega$,

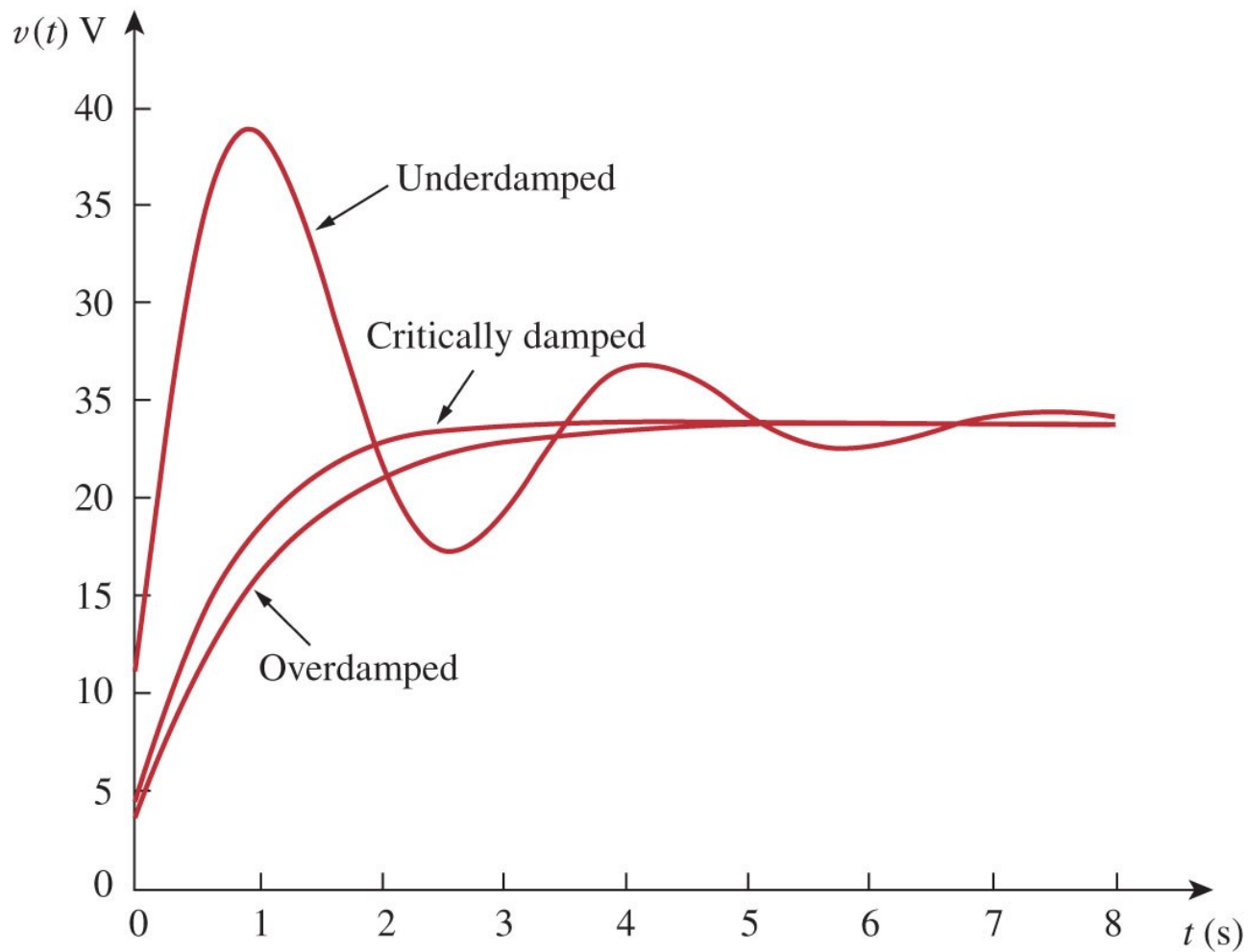
- For $t < 0$, switch closed, capacitor open, inductor shorted.
- For $t > 0$, switch open, a series RLC network
- $v(0) = 12V$, $i(0) = 12A = C \frac{dv(0)}{dt}$, $\frac{dv(0)}{dt} = 48$

$$\alpha = \frac{R}{2L} = 0.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 2, \quad s_{1,2} = -0.5 \pm j1.936 \quad \text{Underdamped}$$

$$v(t) = v_s + (B_1 \cos 1.936t + B_2 \sin 1.936t)e^{-0.5t}$$



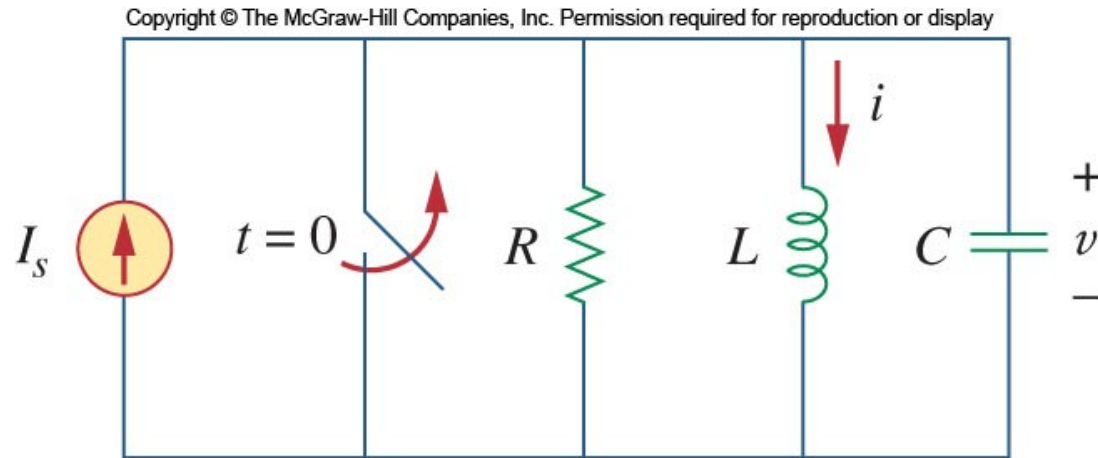
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Step Response of a Parallel RLC Circuit



Apply KCL,

$$\frac{v}{R} + i + C \frac{dv}{dt} = I_s$$

$$\& \quad v = L \frac{di}{dt}$$

So we get

$$\frac{d^2 i}{d^2 t} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$



Step Response of a Parallel RLC Circuit

$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$

- The total response is a combination of **steady state responses and transient response**:

$$i(t) = I_s + (A_1 e^{s_1 t} + A_2 e^{s_2 t}) \text{ (Overdamped)}$$

$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \text{ (Critically Damped)}$$

$$i(t) = I_s + (B_1 \cos \omega_d t + B_2 \sin \omega_d t) e^{-\alpha t} \text{ (Underdamped)}$$

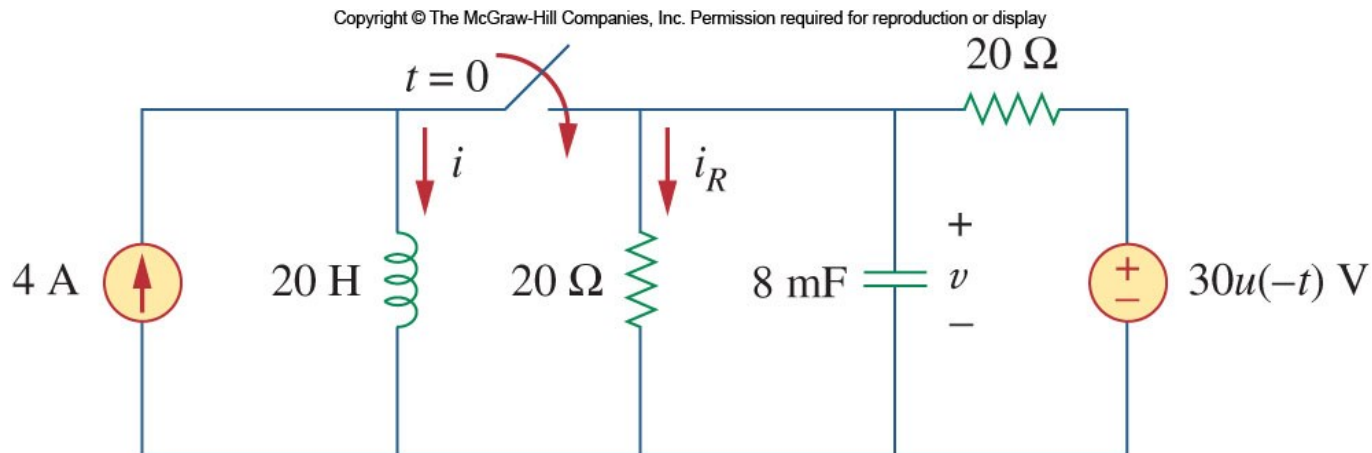
Here the variables A_1/A_2 B_1/B_2 are obtained from the initial conditions, $i(0)$ and $di(0)/dt$.





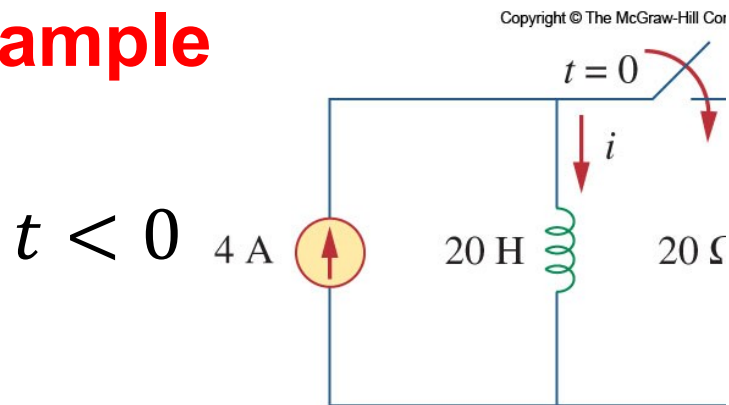
Example

- Find $i(t)$ and $i_R(t)$ for $t > 0$.

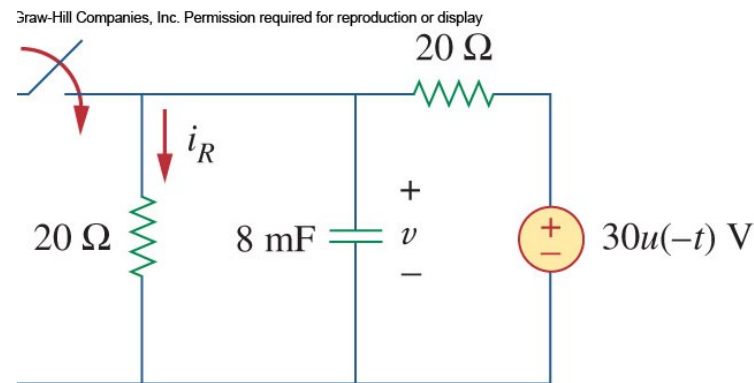




Example

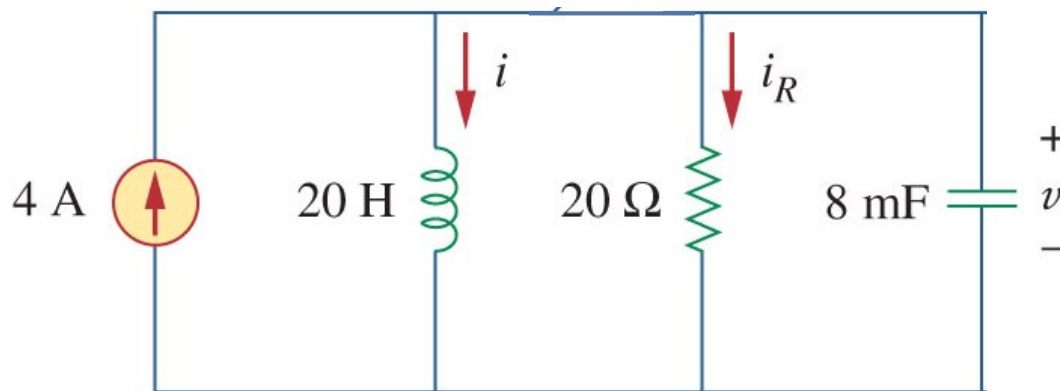


Initial values : $i(0^+) = 4A$



$$v(0^+) = \frac{20}{20 + 20} \times 30V = 15V$$

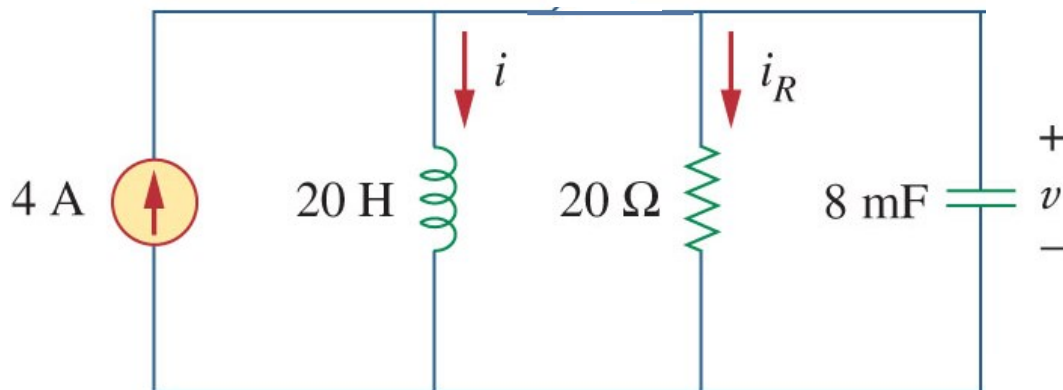
$t = 0^+$



$$v(0^+) = 15V = L \frac{di(0^+)}{dt}$$



$$t = 0^+$$



$$\text{For } t > 0, \alpha = \frac{1}{2RC} = \frac{1}{2 \times 10 \times 8 \times 10^{-3}} = 6.25, \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20 \times 8 \times 10^{-3}}} = 2.5$$

$$s_{1,2} = -6.25 \pm 5.7282$$

$$i(t) = I_{s.s.} + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$i(0^+) = 4A$$

Apply

$$\frac{di(0^+)}{dt}$$

